

0.1 Finite Field

We know that $\mathbb{Z}_p$ is a field where p is prime number. But, there are finite fields of not prime order.

Proposition 0.1.1. Let E be a finite extension field of $\mathbb{Z}_p$. Then E has order of p^n for some $n \in \mathbb{N}$.

Proof. Trivial. □